ZOLEX

Purchase-to-Store Enterprise ERP

Master Software Requirements Specification (SRS) + System Requirement & Development Documentation (SRC)

Company: Zeeta Electrical Engineering Pvt. Ltd.

Brand: Zolex

Technology Target: React + TypeScript + Node.js + SQL

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Purpose: This document is the authoritative product specification for designing, developing, testing and validating a premium enterprise Purchase-to-Store ERP module.

Design ambition: The product must visually and functionally communicate the quality of a large enterprise ERP while remaining focused on the Purchase-to-Store domain.

1. Executive Summary

Zolex Purchase-to-Store ERP is an enterprise procurement and store workflow platform covering master data, purchase indent, approval, purchase order, goods receipt, administration and management analytics. The platform will use SQL as the primary transactional database, with Excel/CSV for interchange and Google Sheets as a temporary synchronization source during the transition/testing period.

The system is intended to be modular, auditable, secure, highly interactive and scalable. Every major operational screen must support filtering, sorting, pagination, export/print, drill-down and clickable details. Analytics must not be decorative: KPI cards, graphs and tables must lead users to underlying transactions.

2. Product Vision

- Create a premium enterprise procurement ERP experience under the Zolex brand.
- Centralize Item, Supplier, Indent, PO and GRN information.
- Reduce manual data entry through master-driven auto-population.
- Provide controlled approval and role-based access.
- Enable management to move from KPI to transaction-level detail.
- Allow Google Sheets to act as a temporary live data bridge without making it the permanent system of record.
- Maintain complete auditability of critical actions.
- Provide Excel/CSV import and export and print-ready reports.

3. Scope

3.1 In Scope

- Management Analysis
- Master Data: Supplier Master and Item Master
- Purchase Indent: Data and Analysis
- Purchase Order: Data and Analysis, Pending PO and Complete PO
- GRN: Data and Analysis
- Direct PO and Indent-based PO
- Multiple-indent consolidation into one PO
- Supplier-wise pending-indent consolidation
- Approval/Pending/Rejected workflow
- Admin: users, roles, permissions, data management, CRUD, import/export and backup
- Google Sheets synchronization
- SQL database
- Universal filtering, drill-down, export and print

3.2 Out of Scope

- Accounting/General Ledger
- Sales/CRM
- Manufacturing/MRP
- Payroll/HR

- Full warehouse management beyond the defined GRN/store receipt scope
- Automatic final academic/financial approval outside configured workflow

4. Navigation & Information Architecture

Primary Area	Sub Areas
Management Analysis	Executive dashboard, KPI cards, charts, analytical tables, drill-down
Master	Supplier Master, Item Master
Purchase Indent	Data, Analysis
Purchase Order	Data, Analysis, Pending PO, Complete PO
GRN	Data, Analysis
Admin	User Management, Role Management, Permission Management, Data Management, Import/Export, Backup, CRUD, Audit

5. User Roles

- Super Admin: unrestricted system administration, security configuration, user/role/permission management, data management and backup.
- Purchase Head: approve/reject purchase indents and perform authorized purchasing analysis.
- Purchase User: create and manage permitted indents and POs.
- Store User: manage authorized GRN/store receipt activities.
- Management: read-only or configured analytical access with drill-down.
- Viewer: read-only access.
- Custom Roles: administrators may create roles with granular permissions.

6. Authentication & Initial Administrator

The initial bootstrap administrator shall use Employee ID **10221** and the supplied initial password **Admin@12345**. This credential is for bootstrap/development use and must be forced to rotate or be changed before production deployment. Authentication must use secure password hashing, session/token controls, logout, failed-login protection and audit logging.

7. Master Data Requirements

7.1 Item Master

Field	Required Behaviour
Item Code	Unique searchable identifier
Item Description	Auto-populated after item selection
Item Type	Master-driven
Department	Master-driven
Cost Centre	Master-driven
Main Group	Master-driven
Sub Main Group	Master-driven
HSN Code	Master-driven
Date / Time	Created/updated metadata

Field	Required Behaviour
Indent Type	Master-driven dropdown where applicable
UOM	Master-driven
Item Specification	Master-driven

7.2 Supplier Master

Field	Required Behaviour
Supplier Name	Unique/searchable supplier
Address	Supplier detail
State / City / Country	Structured location
Mobile No.	Validated contact
Gmail ID	Validated email
GST No	Validated supplier tax identifier
Payment Terms	Supplier commercial information
Currency	Configured currency

7.3 Supporting Masters

- PO Type: dropdown source used wherever PO Type is required.
- Indent Type: dropdown source used wherever Indent Type is required.
- Cost Center: dropdown source used wherever Cost Centre is required.
- User Master: Employee ID, Employee Name and Designation; used as the base user identity source.

8. Google Sheets Integration

Google Sheets will initially act as an external source for data verification and migration support. The application header must contain a clear Live Data Sync control.

- Sync ON: configured sheets are synchronized according to the configured schedule/event mechanism.
- Sync OFF: synchronization is stopped and the ERP continues using its application database.
- Sync must never silently overwrite protected/approved transactional records.
- Every synchronization run must produce status, timestamp, row counts, validation errors and audit information.
- Conflict handling must be deterministic and visible.
- Google Sheet data must be normalized into SQL entities rather than queried directly by every screen.
- Once the organization moves fully to ERP operation, synchronization can be disabled without redesigning the application.

9. Google Sheet Field Mapping

Sheet	Columns / Purpose
Item Master	Item Code, Item Description, Item Type, Department, Cost Centre, Main Group, Sub Main Group, HSN Code, date, time, Indent T
Supplier Master	Supplier Name, Address, State, City, Country, Mobile No., Gmail ID, GST No, Payment Terms, Currency
Purchase Indent	Indent No, date, time, Indent Type, Item code, Item Description, Indent Qty, UOM, Item Specification, Department, Cost Centre, L
Purchase Order	Indent No, Indent type, Indent qty, PO type, date, time, PO no, PO qty, Item code, Item Description, Rate/pc, Discount %, Final Ra

Sheet	Columns / Purpose
GRN	PO no, PO type, Item code, Item description, GRN qty, date, time, Pending GRN qty
PO Type	PO Type
User Master	Employee ID, Employee Name, Designation
Indent Type	Indent Type
Cost Center	Indent Type column as supplied; implementation must confirm whether this is a source-data naming error and preserve a configur

10. Purchase Indent

Purchase Indent must use a header/detail structure. Header information is entered once; item lines are added one by one. Selecting an Item Code must automatically populate all available item master information.

- Create, edit, save draft, submit, view, filter, export and print.
- Indent number generated/validated according to configurable numbering rules.
- Item selection via searchable dropdown.
- Auto-populate description, type, department, cost centre, UOM, specification and other mapped fields.
- Display live quantity and pending GRN quantity where applicable.
- Approval states: Draft, Pending Approval, Approved, Rejected, Cancelled/Closed where configured.
- Purchase Head controls approval/rejection.
- Rejected transactions must retain rejection reason and audit trail.
- Approved transactions must be protected from unauthorized mutation.

11. Purchase Order

11.1 PO Types

- Direct PO: created without an originating indent.
- Indent-based PO: created from eligible approved indents.
- Consolidated PO: multiple eligible indents may be selected and combined into one PO.
- Supplier-wise consolidation: multiple pending indents for the same supplier may be consolidated subject to business rules.

11.2 PO Data

- Indent No / source references
- Indent Type
- Indent Qty
- PO Type
- PO Date / Time
- PO Number
- PO Qty
- Item Code
- Item Description
- Rate per piece

- Discount %
- Final Rate per piece
- Amount
- Item Specification
- Supplier and commercial information

11.3 Quantity Controls

- PO quantity must not exceed the eligible source quantity unless an explicitly authorized over-order rule exists.
- A consolidated PO must retain source-indent traceability for every line.
- Partial PO creation must leave the correct remaining eligible quantity.
- Completed PO lines must not be silently reused.

11.4 Pending vs Complete

Pending PO displays open quantities/commitments. Complete PO displays fully processed/closed POs according to configurable completion rules. Both views must support filters, drill-down, export and print.

12. GRN

- GRN must be traceable to PO and item lines.
- Display PO quantity, received quantity and pending GRN quantity.
- Support partial receipt and final receipt.
- Prevent GRN quantity from exceeding the permitted outstanding PO quantity unless an authorized exception workflow exists.
- Maintain receipt date/time and user audit information.
- GRN completion must update relevant pending balances transactionally.
- GRN Data and Analysis views must support filtering, drill-down, export and print.

13. Management Analysis

The management dashboard is a decision-support layer, not a decorative landing screen.

- Purchase value and quantity KPIs
- Indent count/value and ageing
- Pending approval analysis
- PO count/value and pending value
- GRN quantity/value and pending receipt
- Supplier-wise purchase analysis
- Supplier concentration and performance
- Item-wise purchase analysis
- Department-wise analysis
- Cost Centre-wise analysis
- Indent Type and PO Type analysis
- Date-wise/month-wise/year-wise trends
- Rate and discount analysis

- Pending ageing buckets
- Exception and variance analysis
- Top/bottom suppliers and items
- Purchase cycle-time analysis
- Interactive KPI cards, graphs and analytical tables

14. Universal Filter & Drill-Down Standard

Every major list and analytical component must support relevant combinations of date range, supplier, item, item group, department, cost centre, indent type, PO type, status, employee/user and other applicable dimensions.

- Filters must update related cards, charts and tables coherently.
- Users can click KPI cards, chart segments, table rows and other supported visual elements.
- Clicking opens a modal/drawer/detail page with full contextual information.
- Drill-down must preserve applied filters and source context.
- Users can navigate from management KPI → document → line item → related transaction.
- Every major table supports search, sort, pagination, column control and export.

15. UI/UX & Visual Design System

The target is a premium enterprise visual language inspired by high-end ERP products, without copying proprietary interfaces.

- Premium landing page for Zolex.
- Professional login screen with brand identity.
- Enterprise dashboard shell with sidebar, top header, breadcrumbs and context actions.
- Strong typography hierarchy and consistent spacing system.
- Modern cards, gradients, borders, shadows and controlled animations.
- Rich but disciplined colour system for statuses, alerts and business states.
- Professional iconography and Zolex branding.
- Hover, focus, pressed, loading and disabled states.
- Skeleton loading and meaningful empty states.
- Responsive layouts for desktop and tablet.
- Accessible contrast, keyboard navigation and visible focus states.
- Tables must remain readable with large datasets.
- Animations must support hierarchy and feedback, not reduce performance.

16. Landing Page

- Zolex brand presentation
- Premium hero section
- Enterprise procurement value proposition
- Feature highlights
- Management analytics preview

- Workflow preview
- Security/trust messaging
- Professional CTA to login
- Responsive design

17. Admin Module

- User management: create, edit, activate/deactivate and inspect users.
- Role management: create and modify roles.
- Permission management: granular permissions.
- User-role assignment.
- Data management and controlled CRUD.
- Excel/CSV import with preview and validation.
- Excel/CSV export with filter-aware exports.
- Backup and restore controls.
- Audit log viewer.
- Master data maintenance.
- Configuration management.

18. RBAC Permission Model

Permission	Examples
View	View masters, transactions, analytics
Create	Create indent, PO, GRN, master
Edit	Edit permitted records
Delete	Controlled deletion/soft deletion
Approve	Approve indent or configured workflow
Reject	Reject with mandatory reason
Export	Excel/CSV/PDF/print
Import	Import validated master/transaction data
Analyze	Access analytics and drill-down
Admin	Manage users, roles, permissions, backups and system configuration

19. Database Architecture

SQL is the primary system of record. The database must be located outside the project source folder. The implementation should use normalized transactional entities with foreign keys, unique constraints, indexes, timestamps and audit fields.

- Users / Employees
- Roles
- Permissions
- Role-Permissions

- User-Roles
- Items
- Suppliers
- PO Types
- Indent Types
- Cost Centres
- Purchase Indent Headers
- Purchase Indent Lines
- Indent Approval History
- Purchase Order Headers
- Purchase Order Lines
- PO Source References
- GRN Headers
- GRN Lines
- Google Sync Jobs
- Google Sync Errors
- Import Jobs
- Import Rows/Errors
- Audit Logs
- Backup Metadata

20. Data Integrity Rules

- Use primary and foreign keys.
- Use unique business identifiers where applicable.
- Use soft delete for master records that have historical dependencies.
- Do not hard-delete posted transactional records through normal UI.
- Use database transactions for multi-line document operations.
- Maintain createdBy, createdAt, updatedBy and updatedAt.
- Maintain document status and lifecycle history.
- Validate all foreign-key/master references on both frontend and backend.
- Backend validation is authoritative.

21. API Architecture

- Versioned REST API structure.
- Authentication and authorization middleware.
- Validation middleware.
- Centralized error handling.
- Consistent response envelopes.
- Pagination/filter/sort query standards.

- Idempotency where duplicate submission is possible.
- Audit hooks for critical mutations.
- Rate limiting and abuse protection.
- API documentation and health endpoint.

22. Import / Export

- Upload Excel/CSV.
- Preview rows before commit.
- Validate required fields, master references and duplicates.
- Show valid/invalid counts.
- Allow downloadable error report.
- Commit only validated rows or use explicit partial-import mode.
- Export filtered data.
- Export current view.
- Export full authorized dataset.
- Print-ready reports.

23. Backup & Recovery

- Scheduled database backups.
- Manual backup trigger for authorized administrators.
- Backup metadata and verification status.
- Restore must be protected and auditable.
- Never expose raw database credentials in frontend code.
- Backup files must be stored outside source code and protected by OS/storage permissions.

24. Audit Logging

Critical events must record actor, timestamp, action, entity, entity ID, previous value where appropriate, new value where appropriate, source, request/session information and result.

- Login/logout
- User/role/permission changes
- Master CRUD
- Indent submission/approval/rejection
- PO creation/edit/closure
- GRN creation/edit/closure
- Imports
- Exports
- Google synchronization
- Backup/restore
- Administrative changes

25. Security Requirements

- Password hashes only; never store plaintext passwords.
- Secure authentication tokens/session handling.
- RBAC enforced server-side.
- Input validation and sanitization.
- Parameterized SQL/ORM queries.
- Protection against injection, XSS, CSRF where applicable.
- Secure HTTP headers.
- Secrets only through environment/configuration outside source control.
- Audit sensitive actions.
- Production credentials must be rotated from development defaults.

26. Project Structure

The application must have exactly two top-level application folders:

```
frontend/
React + TypeScript application

backend/
Node.js + API + database integration
```

The SQL database and persistent backup/data directories must be outside the project folder.

Development commands:

```
cd backend
npm install
npm run dev

cd frontend
npm install
npm run dev
```

27. Environment Configuration

- Backend environment file/config contains database URL, authentication secrets, Google integration credentials, storage paths and other secrets.
- Frontend must receive only non-sensitive public configuration.
- Secrets must never be committed to Git.
- Environment validation must fail fast when required configuration is absent.

28. Testing Strategy

- Unit tests for business rules and calculations.
- Integration tests for API/database workflows.
- End-to-end tests for Indent → Approval → PO → GRN.
- RBAC tests for each major permission.
- Import/export tests.
- Google synchronization tests.

- Analytics correctness tests against known fixtures.
- Quantity-balance tests.
- Regression tests after fixes.
- Security tests.
- Performance tests with realistic data volumes.
- Browser compatibility and responsive UI tests.

29. Acceptance Criteria

- Project starts successfully using the two documented npm commands.
- Landing and login work.
- Super Admin can access authorized administration.
- All masters support CRUD according to permissions.
- Dropdowns are master-driven.
- Item and supplier selection auto-populates mapped information.
- Indent approval works.
- Direct and indent-based PO work.
- Multiple eligible indents can be consolidated into a PO.
- PO pending/complete logic works.
- GRN supports partial and complete receipt.
- Pending quantities remain mathematically correct.
- Management analytics reconcile with transactional data.
- Filters update all relevant components.
- Drill-down reaches source transactions.
- Excel/CSV import/export works.
- Print works.
- Google sync ON/OFF works safely.
- Audit logs exist for critical mutations.
- No unauthorized user can bypass backend permissions.
- Automated tests pass before release.

30. Performance & Scalability Targets

- Use indexed filtering for common dimensions.
- Server-side pagination for large datasets.
- Server-side sorting and filtering.
- Efficient aggregate queries for dashboards.
- Cache only where correctness is preserved.
- Bulk operations for imports.
- Avoid N+1 queries.
- Use connection pooling.

- Use asynchronous jobs for long-running synchronization/import/export operations.
- Analytics must not block transactional operations.

31. Development Governance

- Documentation is the source of truth.
- AI agent must inspect the existing workspace before modifying it.
- Do not blindly overwrite existing files.
- Do not create duplicate implementations of the same feature.
- Maintain an implementation status document.
- Maintain changelog and known-issues log.
- Run relevant tests after each meaningful implementation stage.
- Fix discovered build/test/runtime issues before proceeding.
- Use deterministic naming and folder conventions.
- Do not stop for trivial decisions; use the SRS and engineering judgement.
- Do not claim completion without verifying actual code, database and UI behaviour.

32. Autonomous AI Development Protocol

The eventual Anthropic development prompt must instruct the agent to operate as a coordinated enterprise engineering team: product architect, business analyst, UX architect, frontend engineer, backend engineer, SQL architect, security engineer, QA engineer, DevOps engineer and code reviewer.

- Phase 0: inspect repository/workspace and produce an inventory.
- Phase 1: establish documentation and architecture.
- Phase 2: establish database and backend foundation.
- Phase 3: establish design system and application shell.
- Phase 4: implement masters.
- Phase 5: implement indent.
- Phase 6: implement PO.
- Phase 7: implement GRN.
- Phase 8: implement analytics.
- Phase 9: implement Google sync.
- Phase 10: implement Admin/import/export/backup/audit.
- Phase 11: perform full QA and security validation.
- Phase 12: perform final forensic release audit.

33. Implementation Status Discipline

The agent must maintain a machine-readable and human-readable status such as: NOT_STARTED, IN_PROGRESS, BLOCKED, IMPLEMENTED, TESTED, VERIFIED. A feature is not COMPLETE merely because code exists; its acceptance criteria, tests and runtime behaviour must be verified.

34. Business Rules Summary

- Master data drives transactional dropdowns.
- Source references must be retained for consolidated POs.
- Approved/rejected state changes require authorized roles.
- PO balances derive from ordered quantities and valid source references.
- GRN balances derive from PO quantities and receipts.
- Transaction histories are immutable or controlled through audited correction workflows.
- Analytics must use the same authoritative transaction definitions as operational screens.

35. Reporting Requirements

- Every operational table must support print/export.
- Reports must respect active filters.
- Export must preserve business-relevant fields and human-readable headers.
- Print layouts must include Zolex/Zeeta branding, report title, filter context, generated timestamp and page numbering.
- Management dashboards must offer snapshot/export capability.

36. UX Interaction Standards

- Use confirmation for destructive/irreversible actions.
- Use clear status colours and text, not colour alone.
- Show loading state during network operations.
- Disable duplicate submissions while a request is processing.
- Show actionable validation errors next to fields.
- Provide success/error feedback.
- Preserve unsaved data warnings where relevant.
- Allow keyboard-friendly table and form interaction.

37. Recommended Entity Relationships

Supplier 1→N Purchase Orders; Item 1→N Indent Lines; Item 1→N PO Lines; Item 1→N GRN Lines; Indent Header 1→N Indent Lines; PO Header 1→N PO Lines; PO Lines may reference one or more originating indent lines through a source-reference entity; PO Header 1→N GRN Headers/Lines as permitted by the document model.

38. Analytics Calculation Governance

- Define each KPI in a central metric catalogue.
- Use consistent inclusion/exclusion rules.
- Do not calculate a KPI differently on different screens.
- Show data freshness.
- Show active filters.
- Allow drill-down to source records.
- Handle null/unknown values explicitly.
- Document all derived metrics.

39. Migration Strategy

- Validate Google Sheet structure.
- Map columns to canonical SQL entities.
- Import masters first.
- Validate duplicate and missing references.
- Import historical transactional data in dependency order.
- Reconcile row counts and key totals.
- Run business-rule validation.
- Generate migration exception report.
- Only then enable ERP operational workflows.

40. Release Gates

- G0 Documentation approved
- G1 Architecture/database verified
- G2 Authentication/RBAC verified
- G3 Masters verified
- G4 Indent workflow verified
- G5 PO workflow verified
- G6 GRN workflow verified
- G7 Analytics reconciliation verified
- G8 Sync/import/export verified
- G9 Security/performance verified
- G10 UAT and final release audit passed

41. Traceability Matrix Template

Requirement	UI	API	DB	Business Rule	Test	Status
Supplier Master	Supplier UI	Supplier API	Supplier	Master validation	Supplier CRUD tests	To implement
Item Master	Item UI	Item API	Item	Auto-population	Item CRUD tests	To implement
Indent	Indent UI	Indent API	Indent Header/Line	Approval rules	E2E indent tests	To implement
PO	PO UI	PO API	PO Header/Line	Source/quantity rules	E2E PO tests	To implement
GRN	GRN UI	GRN API	GRN Header/Line	Receipt balance	E2E GRN tests	To implement
Analytics	Dashboard	Analytics API	Aggregates	Metric definitions	Reconciliation tests	To implement
Google Sync	Header control/Admin	Sync API	Sync Jobs	Conflict rules	Sync tests	To implement

42. Final Product Quality Definition

The finished product is considered enterprise-grade only when visual polish, functional completeness, data integrity, security, auditability, workflow correctness, analytics correctness and operational reliability are all demonstrated together. A visually attractive frontend without trustworthy transaction logic is not acceptable; likewise, technically correct CRUD without a premium enterprise UX does not meet the product vision.

43. Final Autonomous Development Sequence

The final implementation prompt should require the AI agent to execute: Inspect → Document → Architect → Database → Backend foundation → Frontend shell → Masters → Indent → Approval → PO → GRN → Analytics → Google Sync → Admin → Import/Export → Backup/Audit → Testing → Security → Performance → Reconciliation → Final UI/UX polish → Release audit.

The agent must continue autonomously through non-blocking issues, create/modify files as needed, run commands/tests, inspect failures and correct them. It must not report a feature as complete without verification.

44. Important Open Decisions to Confirm During Implementation

- Exact SQL engine/version (PostgreSQL recommended unless company standard dictates otherwise).
- Google Sheets synchronization frequency/event strategy and service-account ownership.
- Exact PO completion definition.
- Whether Supplier Master contains supplier code in addition to supplier name.
- Whether Cost Center source column really is named 'Indent Type' or is a sheet-header typo.
- Whether GRN has a separate GRN Number requirement.
- Currency and tax calculation requirements beyond the supplied fields.
- Document numbering format for Indent, PO and GRN.
- Exact approval hierarchy beyond Purchase Head.
- Backup retention period and storage location.

45. Deliverables Expected From the AI Development Agent

- Working frontend application
- Working backend API
- SQL schema/migrations
- Seed/bootstrap scripts
- Authentication/RBAC
- All specified modules
- Google Sheets integration
- Import/export
- Backup functionality
- Audit logs
- Automated tests
- API documentation
- Deployment/configuration documentation
- Implementation status
- Changelog
- Known issues
- Final release-readiness report

46. Document Control

Property	Value
Product	Zolex Purchase-to-Store Enterprise ERP
Company	Zeeta Electrical Engineering Pvt. Ltd.
Document	Master SRS + SRC
Version	1.0
Date	24 August 2026
Primary Stack	React + TypeScript + Node.js + SQL
Primary Data Store	SQL
External Bridge	Google Sheets
Interchange	Excel / CSV
Development Model	Autonomous AI-assisted enterprise development